

EXP-2026-0001 — Testing Exo7

Owner: RJ | **Project:** ATLAS | **Prepared:** 2026-09-11 | **Status:** READY_TO_RUN

Bench sheet generated from `experiments/2026/EXP-2026-0001/manifest.yaml`. The manifest remains the authoritative record; this sheet is the working copy for the person running the experiment.

1. What this experiment tests

Exo7 degrades single-stranded DNA from free ends but does not act on double-stranded DNA. A correctly hybridized padlock (circular) encoding probe has its 5' and 3' ends held in duplex, so it should be protected. A probe bound incorrectly leaves exposed ssDNA ends, so it should be digested.

To test this without relying on genuinely non-specific binding, a **linear** rRNA probe is used as a stand-in for the exposed-end case and a **circular** rRNA probe for the protected case. Both bind rRNA specifically; they differ in end structure.

Measurement: cytoplasmic fluorescence intensity per cell.

A positive result is Exo7 substantially reducing linear-probe signal while circular-probe signal stays largely intact.

Read this before you start

The linear probe is doing double duty — it is both the test arm and the only check that the Exo7 enzyme worked at all. If the linear signal does **not** drop, you cannot tell whether the enzyme was inactive or the hypothesis is wrong. If that happens, do not discard the run; flag it, because distinguishing those two cases needs a separate enzyme-activity control.

2. Conditions and plate layout

Ten wells, one plate, one plating batch. The two wells per condition are **technical** replicates.

Cond	Wells	Encoding probe	Exo7	Readout probe
C1	A1, A2	linear-to-rRNA	+	YEH1
C2	A3, A4	linear-to-rRNA	– (mock)	YEH1
C3	A5, A6	circular-to-rRNA	+	YEH2
C4	A7, A8	circular-to-rRNA	– (mock)	YEH2
C5	A9	none	– (mock)	YEH1
C6	A10	none	– (mock)	none

C5 is the background for YEH1 binding without a target-specific encoder. C6 is bare cell/plate autofluorescence.

The well assignment above is a proposal — the manifest does not fix physical wells. Use it or change it, but write down what you actually used.

3. Reagents and stocks

Item	Stock	Used for
Poly-D-lysine	0.1 mg/mL	plate coating
3T3 cells	T75 flask	plating
Trypsin-EDTA	0.5%	harvest
PFA	32%	fixation
PBS, PBST + PVSA	—	washes, storage
Linear rRNA encoding probe	100 nM	C1, C2
Circular rRNA encoding probe	100 nM	C3, C4
YEH1 readout probe	100 nM	C1, C2, C5
YEH2 readout probe	100 nM	C3, C4
Formamide	50%	both hybridizations
TBST	—	hybridization + washes
Exo7	<i>record U/μL from the tube</i>	C1, C3
TBE	1× standard	Exo7 + mock

4. Day 1 — plate, fix, and start encoding hybridization

Protocol: [pdl-coating-cell-plating-fixation](#) v0.1. All volumes below include **10% overage**.

4a. Coat the plate

Component	Per well	Total (10 wells + 10%)
PDL, 0.1 mg/mL	25 μL	275 μL

1. Add 25 μL PDL per well. Incubate **1 h at room temperature**.
2. Wash 3× with PBS, enough to cover the surface each time.
3. Dry uncovered in the biosafety cabinet, **1 h**.

4b. Harvest, count, and plate

Target is **10,000 cells per well in 50 μL**, i.e. a plating suspension at **200 cells/μL**.

Quantity	Value
Total plating suspension needed	550 μL
Total cells needed	110,000
Required suspension concentration	200 cells/μL

1. Trypsinize with 2.5 mL 0.5% trypsin-EDTA per T75; detach and collect per the usual 3T3 procedure; resuspend in complete medium.

- Count cells and record the concentration below. **This run uses a total cell count, not a viability-gated count** — a deliberate deviation from the protocol, which specifies viable cells.

- Compute and record:

Fill in	Value
Measured concentration (cells/ μL)	_____
Cell suspension volume = $110,000 \div$ measured conc.	_____ μL
Medium to add = $550 -$ suspension volume	_____ μL

If the measured suspension is below 200 cells/ μL it cannot reach target by dilution — concentrate, recount, recompute.

- Mix gently and plate 50 μL per well into all 10 wells.

4c. Attachment period

Culture the cells before fixation. **Duration is not specified in the manifest or the protocol — confirm with RJ before starting.**

4d. Fix

Component	Per well	Total (10 wells + 10%)
PFA, 32%	7.14 μL	78.5 μL

- Add 7.14 μL of 32% PFA directly into the 50 μL of medium in each well $\rightarrow$ 4% PFA in ~ 57 μL . **Fume hood, PFA is toxic.**
- Incubate **20 min at room temperature.**
- Dispose of fixative as hazardous waste.
- Wash **3 $\times$ 5 min** with PBST + PVSA.

No separate permeabilization step — this fixation and wash serves that purpose.

4e. Pre-hybridization washes and encoding-probe hybridization

Protocol: [encoding-hybridization-imaging v0.2](#), steps 1–2.

- Wash **3 $\times$ 5 min at 37 °C** in 20% formamide / TBST.
- Prepare the three encoding mixes below. Each well receives **40 μL** ; final probe concentration is **1 nM**, final formamide **20%**.

Mix	Wells	Probe stock (100 nM)	Formamide 50%	TBST	Total
Linear	A1–A4 (4)	1.76 μL	70.4 μL	103.8 μL	176 μL
Circular	A5–A8 (4)	1.76 μL	70.4 μL	103.8 μL	176 μL
No-probe	A9, A10 (2)	—	35.2 μL	52.8 μL	88 μL

Per well that works out to 0.4 μL probe stock + 16 μL formamide + 23.6 μL TBST (24 μL TBST in the no-probe mix).

- Add 40 μL of the appropriate mix to each well.
- Incubate **overnight at 37 °C.**

5. Day 2 — Exo7, readout, imaging

Total hands-on plus incubation is roughly **4.5 h before imaging**.

5a. Post-hybridization wash

Wash **2 × 10 min**. Buffer and temperature are assumed TBST at room temperature — **not explicitly confirmed; check with RJ**.

5b. Exo7 / mock incubation

This is the experimental variable. It is a run-specific step, not a canonical protocol. **50 µL per well, 10 U Exo7 per treated well, 1× TBE, 3 h at 37 °C**.

All ten wells are on one plate and **must be incubated together for the same 3 h** — the mock wells get buffer without enzyme so that the incubation itself is not a confound.

Mix	Wells	Exo7	TBE	Total
+Exo7	A1, A2, A5, A6 (4)	44 U total	to 220 µL	220 µL
Mock	A3, A4, A7, A8, A9, A10 (6)	none	330 µL	330 µL

The Exo7 **volume** depends on the tube's activity, which is not recorded in the manifest:

Fill in	Value
Exo7 stock activity (U/µL)	_____
Exo7 volume = 44 U ÷ stock activity	_____ µL
TBE = 220 µL – Exo7 volume	_____ µL

1. Add 50 µL of the appropriate mix to each well.
2. Incubate **3 h at 37 °C**, all wells together.

5c. Wash

Wash **2 × 10 min** (TBST, room temperature — same assumption as 5a).

5d. Readout-probe hybridization

Protocol: **encoding-hybridization-imaging** v0.2, step 4. **50 µL per well, 5 nM final probe, 10% final formamide, 30 min at room temperature**.

Mix	Wells	Probe stock (100 nM)	Formamide 50%	TBST	Total
YEH 1	A1–A4, A9 (5)	13.75 µL	55.0 µL	206.3 µL	275 µL
YEH 2	A5–A8 (4)	11.0 µL	44.0 µL	165.0 µL	220 µL

Per well: 2.5 µL probe stock + 10 µL formamide + 37.5 µL TBST.

A10 (C6) gets no readout probe. Whether it should still receive 50 µL of probe-free buffer for the same 30 min — matching the handling as the Exo7 mock does — is not specified. Confirm with RJ; see the checklist in section 7.

5e. Final wash and imaging

- 1. Wash **2 × 10 min** (TBST, room temperature — same assumption as 5a).
- 2. Image. **Acquisition settings are deliberately left to you at the scope** — set exposure and channels after checking signal, then record what you used in the table below.

Setting	Value used
Microscope / objective	_____
Channel(s) and fluorophore	_____
Exposure time	_____
Illumination power	_____
Fields per well	_____
Z-range / step	_____
Pixel size / binning	_____
Output dataset ID	_____

Keep settings **identical across all ten wells** — the comparison is between wells, so any per-well change in exposure or power invalidates it.

6. What gets measured

Primary measurement is **cytoplasmic intensity per cell**. The planned comparisons are +Exo7 vs –Exo7 within each probe type (C1 vs C2, C3 vs C4), then the relative change between probe types. C5 and C6 serve as background references, not part of the primary comparison.

Segmentation, background subtraction, and QC exclusions are intentionally not fixed in advance — they will be chosen at analysis time.

7. Confirm before you start

- 1. Attachment/growth period between plating and fixation — not specified anywhere.
- 2. Wash buffer and temperature for the three 2 × 10 min washes — assumed TBST at room temperature, never confirmed.
- 3. Exo7 stock activity in U/μL, to convert 44 U into a volume.
- 4. Whether A10 (C6) receives probe-free buffer during the readout step.
- 5. Physical well assignment — confirm or replace the proposed A1–A10 map.

8. Known limitations of this design

These are recorded in the manifest and are accepted, not oversights:

1. **n = 2 technical replicates from one plating batch.** No biological replication; this is a pilot, not a quantitative result.
2. **The linear probe is both the test arm and the only Exo7 activity control** — a null result will be ambiguous.
3. **Only YEH1 has a background well (C5).** If YEH1 and YEH2 differ in non-specific binding, C3/C4 have no matched background.
4. **C6 measures general autofluorescence,** not whether the encoding probe itself fluoresces without a readout probe attached. Nothing in this design tests the latter.
5. **Cell counts are total, not viability-gated,** deviating from the plating protocol.